

The scalar two-point function in de Sitter space

Bunch–Davies normalization and input to the scalar-QED loop

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Abstract

We construct the free charged-scalar two-point function needed for the expansion of the de Sitter scalar-QED effective action about zero vector background. The planar-patch modes are normalized for both real and imaginary Hankel order, and their ultraviolet limit is compared with the local Minkowski singularity. The invariant hypergeometric form is identified using a known evaluation of the de Sitter mode sum; a direct proof of the Hankel transform is not supplied here. We formulate this identification as an equality of distributional boundary values, establish its infrared domain, and check the Klein–Gordon equation, the short-distance expansion and the conformal limit. The final section gives the closed-time-path components, their contact equations and the derivatives required by the polarization calculation.

1 Question and conventions

The scalar part of the programme in Ref. [1] is to obtain the exact free two-point function in de Sitter space, first checking its sub-horizon normalization and then recovering its invariant form. This is the matter input to the one-loop polarization discussed in Ref. [2]. Throughout this note the classical vector background is set to $A_\mu = 0$. The result therefore describes the expansion about zero field; it is not the scalar propagator in a finite electric background.

We work in the expanding planar patch with signature $(-, +, +, +)$,

$$ds^2 = a^2(\eta)(-d\eta^2 + d\mathbf{x}^2), \quad a(\eta) = -\frac{1}{H\eta}, \quad \eta < 0, \quad R = 12H^2. \quad (1)$$

The scalar is complex and has effective mass

$$M_\phi^2 = m_\phi^2 + \xi R = m_\phi^2 + 12\xi H^2. \quad (2)$$

Minimal coupling corresponds to $\xi = 0$; a massless conformally coupled scalar has $m_\phi = 0$, $\xi = 1/6$. The positive-frequency two-point function is defined without an overall factor of i ,

$$G^+(x, x') = \langle 0_{\text{BD}} | \hat{\phi}(x) \hat{\phi}^\dagger(x') | 0_{\text{BD}} \rangle. \quad (3)$$

The dagger matters for a charged field. We use “two-point function” for this Wightman boundary value and distinguish it from a time-ordered Green function when contact terms enter.

The scalar state is Bunch–Davies. For real M_ϕ^2 , we take the principal branch of

$$\nu^2 = \frac{9}{4} - \frac{M_\phi^2}{H^2}. \quad (4)$$

Thus $\nu \geq 0$ is real or $\nu = i\rho$ with $\rho > 0$. The unregulated planar-patch state used below exists for $\text{Re } \nu < 3/2$, equivalently $M_\phi^2 > 0$ in this family. The massless and tachyonic exceptions are discussed in Section 5.3.

2 Modes and canonical normalization

2.1 The classical equation

The free action and its equation of motion are

$$S_\phi = - \int d^4x \sqrt{-g} \left[g^{\mu\nu} (\partial_\mu \phi)^* (\partial_\nu \phi) + M_\phi^2 \phi^* \phi \right], \quad (5)$$

$$(\square - M_\phi^2) \phi = 0, \quad \square \phi = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu \phi). \quad (6)$$

Since $\sqrt{-g} = a^4$ and $g^{\mu\nu} = a^{-2} \eta^{\mu\nu}$,

$$\square \phi = a^{-2} \left[-\phi'' - 2\mathcal{H}\phi' + \nabla^2 \phi \right], \quad \mathcal{H} = \frac{a'}{a} = -\frac{1}{\eta}. \quad (7)$$

A spatial Fourier mode $\phi_k(\eta) e^{i\mathbf{k}\cdot\mathbf{x}}$ therefore obeys

$$\phi_k'' + 2\mathcal{H}\phi_k' + (k^2 + a^2 M_\phi^2) \phi_k = 0. \quad (8)$$

With $v_k = a\phi_k$, the first derivative cancels:

$$v_k'' + \left(k^2 + a^2 M_\phi^2 - \frac{a''}{a} \right) v_k = 0, \quad \frac{a''}{a} = \frac{2}{\eta^2}, \quad (9)$$

$$v_k'' + \left[k^2 - \frac{\nu^2 - \frac{1}{4}}{\eta^2} \right] v_k = 0. \quad (10)$$

This rescaling also makes the short-wavelength normalization transparent.

2.2 The Bunch–Davies solution

For $z = -k\eta > 0$, the general solution is

$$v_k(\eta) = \sqrt{-\eta} \left[A_k H_\nu^{(1)}(-k\eta) + B_k H_\nu^{(2)}(-k\eta) \right]. \quad (11)$$

At fixed order, the large- z asymptotic formula is

$$H_\nu^{(1)}(z) \sim \sqrt{\frac{2}{\pi z}} \exp \left[i \left(z - \frac{\pi\nu}{2} - \frac{\pi}{4} \right) \right]. \quad (12)$$

The condition $v_k \sim e^{-ik\eta}/\sqrt{2k}$ as $\eta \rightarrow -\infty$ selects

$$\boxed{\phi_k^{\text{BD}}(\eta) = \frac{\sqrt{\pi}}{2} H(-\eta)^{3/2} e^{i\pi(2\nu+1)/4} H_\nu^{(1)}(-k\eta).} \quad (13)$$

For real ν , the phase convention in Ref. [1] is related by

$$\phi_k^{\text{project}} = -\frac{\sqrt{\pi}}{2} H(-\eta)^{3/2} H_\nu^{(1)}(-k\eta) = e^{-i\pi(2\nu-3)/4} \phi_k^{\text{BD}}. \quad (14)$$

The phase cancels in G^+ . For imaginary order, however,

$$\left| e^{i\pi(2i\rho+1)/4} \right| = e^{-\pi\rho/2}; \quad (15)$$

this is a normalization factor, not a removable phase. It must be retained when $M_\phi^2/H^2 > 9/4$.

To check the normalization, write $u = -\eta$ and $C_\nu = (\sqrt{\pi}/2)e^{i\pi(2\nu+1)/4}$. The canonical commutator requires

$$a^2 (\phi_k \phi_k^{*'} - \phi_k^* \phi_k') = i. \quad (16)$$

For real order, conjugation interchanges the Hankel functions and $|C_\nu|^2 = \pi/4$. The derivative of $u^{3/2}$ cancels from the antisymmetric product, leaving the Hankel Wronskian,

$$H_\nu^{(1)}(z) \frac{dH_\nu^{(2)}}{dz} - H_\nu^{(2)}(z) \frac{dH_\nu^{(1)}}{dz} = -\frac{4i}{\pi z}, \quad (17)$$

$$\phi_k \phi_k^{*'} - \phi_k^* \phi_k' = -\frac{\pi}{4} H^2 u^3 k \left(-\frac{4i}{\pi k u} \right) = i H^2 u^2 = \frac{i}{a^2}. \quad (18)$$

For $\nu = i\rho$ and real positive z ,

$$[H_{i\rho}^{(1)}(z)]^* = H_{-i\rho}^{(2)}(z) = e^{\pi\rho} H_{i\rho}^{(2)}(z), \quad |C_{i\rho}|^2 = \frac{\pi}{4} e^{-\pi\rho}. \quad (19)$$

The two real factors cancel, so the same Wronskian proof applies. These identities also explain the complex-order normalization in Ref. [5], Eq. (11).

3 The ultraviolet check before the mode integral

Equation (13) gives

$$\phi_k^{\text{BD}}(\eta) \sim \frac{e^{-ik\eta}}{a(\eta)\sqrt{2k}}. \quad (20)$$

For comparison with a massless Minkowski mode, the physical momentum must be large relative to both curvature and mass: $k/a \gg H, M_\phi$. The condition $-k\eta \gg 1$ alone is not a uniform approximation in a parametrically large mass.

At two times, with $a_i = a(\eta_i)$ and $\Delta\eta = \eta_1 - \eta_2$, the high-momentum part of the mode product is

$$\phi_k(\eta_1) \phi_k^*(\eta_2) \sim \frac{e^{-ik\Delta\eta}}{2k a_1 a_2}. \quad (21)$$

Consequently the leading short-distance singularity of G^+ has the representation

$$G^+(x, x') \sim_{\text{UV}} \frac{1}{a_1 a_2} \int \frac{d^3 k}{(2\pi)^3} \frac{e^{-ik\Delta\eta + i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')}}{2k}. \quad (22)$$

The symbol $\sim_{\text{UV}}$ compares singular parts; it does not replace the low-momentum portion of the massive correlator. An intermediate comoving cutoff K satisfying $K/a_i \gg H, M_\phi$ may be introduced to make this separation explicit. Extending the leading ultraviolet integrand to $k = 0$ changes smooth local terms, not its leading coincidence singularity.

There is a useful correction to the physical-momentum check in Ref. [1]. At unequal times there is no unique scale factor with which to define $\mathbf{q} = \mathbf{k}/a$. Choose a reference time $\bar{\eta}$ in a small neighbourhood of the two points and set

$$\mathbf{q} = \frac{\mathbf{k}}{\bar{a}}, \quad \Delta\mathbf{X} = \bar{a}(\mathbf{x} - \mathbf{x}'), \quad \Delta t \simeq \bar{a}\Delta\eta, \quad \bar{a} = a(\bar{\eta}). \quad (23)$$

Then $d^3 k = \bar{a}^3 d^3 q$ and $a_1 a_2 \simeq \bar{a}^2$. All scale factors cancel:

$$G^+(x, x') \sim_{\text{UV}} \int \frac{d^3 q}{(2\pi)^3} \frac{1}{2q} e^{-iq\Delta t + i\mathbf{q} \cdot \Delta\mathbf{X}}. \quad (24)$$

This is the local inertial Minkowski singularity. Keeping a prefactor $(a_1 a_2)^{-1}$ after the change of integration measure, or retaining $q\Delta\eta$ instead of $q\Delta t$, would mix the two coordinate descriptions.

4 The mode sum and its invariant form

4.1 Charged-field contractions

At zero background field, particles and antiparticles have the same mode functions. The operator expansion is

$$\hat{\phi}(\eta, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \left[\phi_k(\eta) e^{i\mathbf{k}\cdot\mathbf{x}} \hat{a}_{\mathbf{k}} + \phi_k^*(\eta) e^{-i\mathbf{k}\cdot\mathbf{x}} \hat{b}_{\mathbf{k}}^\dagger \right], \quad (25)$$

with

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}^\dagger] = [\hat{b}_{\mathbf{k}}, \hat{b}_{\mathbf{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}'), \quad (26)$$

$$\hat{a}_{\mathbf{k}}|0_{\text{BD}}\rangle = \hat{b}_{\mathbf{k}}|0_{\text{BD}}\rangle = 0. \quad (27)$$

Only the $\hat{a}\hat{a}^\dagger$ contraction contributes to Eq. (3). Thus, writing $r = |\mathbf{x} - \mathbf{x}'|$,

$$G^+(x, x') = \int \frac{d^3k}{(2\pi)^3} \phi_k(\eta_1) \phi_k^*(\eta_2) e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')} \quad (28)$$

$$= \frac{1}{2\pi^2} \int_0^\infty dk k^2 \frac{\sin kr}{kr} \phi_k(\eta_1) \phi_k^*(\eta_2), \quad (29)$$

where $\int d\Omega e^{ikr \cos \theta} = 4\pi \sin(kr)/(kr)$.

Set $u = -\eta_1 > 0$, $v = -\eta_2 > 0$. The normalization and order-reflection factors give, for either real or imaginary order,

$$\phi_k(\eta_1) \phi_k^*(\eta_2) = \frac{\pi H^2}{4} (uv)^{3/2} H_\nu^{(1)}(ku) H_\nu^{(2)}(kv). \quad (30)$$

The non-elementary part is therefore the Hankel-product transform

$$G^+(x, x') = \frac{H^2(uv)^{3/2}}{8\pi r} \mathcal{I}_\nu(u, v, r), \quad (31)$$

where the distribution $\mathcal{I}_\nu$ is defined below.

4.2 Regulation and the known transform

The large- k integral is oscillatory and needs a boundary-value prescription. An Abel regulator defines

$$\mathcal{I}_\nu(u, v, r) = \lim_{\epsilon \downarrow 0} \int_0^\infty dk k \sin(kr) H_\nu^{(1)}(ku) H_\nu^{(2)}(kv) e^{-\epsilon k}. \quad (32)$$

The limit is distributional where necessary. For finite $\epsilon > 0$ the ultraviolet endpoint is damped; the infrared condition is $\text{Re } \nu < 3/2$. We initially take $r > 0$ and recover $r = 0$ through the two-point distribution, without assigning a finite value at coincidence.

Define the real invariant and its Wightman boundary value by

$$Z = 1 + \frac{(\eta_1 - \eta_2)^2 - r^2}{2\eta_1 \eta_2} = \frac{\eta_1^2 + \eta_2^2 - r^2}{2\eta_1 \eta_2}, \quad (33)$$

$$y = 2(1 - Z), \quad Z_\delta = 1 - \frac{r^2 - (\Delta\eta - i\delta)^2}{2\eta_1 \eta_2}, \quad \delta > 0. \quad (34)$$

In the limit, $Z_\delta \rightarrow Z - i0 \text{sgn}(\Delta\eta)$. The sign follows from positive frequency: $e^{-ik(\Delta\eta - i\delta)} = e^{-ik\Delta\eta} e^{-k\delta}$. The finite- δ expression is a convenient route to the invariant boundary value; it is not asserted to reproduce the Abel-regulated integral at finite regulator.

Let

$$A = \frac{3}{2} + \nu, \quad B = \frac{3}{2} - \nu, \quad C_\nu = \frac{H^2}{16\pi^2} \Gamma(A) \Gamma(B). \quad (35)$$

The known evaluation of the uncompactified de Sitter mode sum gives

$$G^+(x, x') = C_\nu \lim_{\delta \downarrow 0} {}_2F_1\left(A, B; 2; \frac{1 + Z_\delta}{2}\right). \quad (36)$$

Equation (22) of Ref. [5] expresses the same function in terms of an associated Legendre function, and Eq. (5) of Ref. [6] gives the normalized hypergeometric form directly. For the latter reference the regularized hypergeometric function equals ${}_2F_1$ here because $\Gamma(2) = 1$.

For completeness, the conversion from the Legendre form can be displayed without assuming its normalization. Put $w = -Z$, initially in the real region $w > 1$. In four dimensions, Ref. [5] gives

$$G^+ = \frac{H^2 \Gamma(A) \Gamma(B)}{8\pi^2} \frac{P_{\nu-1/2}^{-1}(w)}{\sqrt{w^2 - 1}}. \quad (37)$$

The representation of P and Euler's hypergeometric transformation imply

$$\begin{aligned} \frac{P_{\nu-1/2}^{-1}(w)}{\sqrt{w^2 - 1}} &= \frac{1}{w + 1} {}_2F_1\left(\frac{1}{2} - \nu, \frac{1}{2} + \nu; 2; \frac{1-w}{2}\right) \\ &= \frac{1}{2} {}_2F_1\left(\frac{3}{2} + \nu, \frac{3}{2} - \nu; 2; \frac{1-w}{2}\right). \end{aligned} \quad (38)$$

Continuation with the Wightman boundary prescription then yields Eq. (36).

Combining this known result with Eq. (31) identifies the transform as

$$\mathcal{I}_\nu(u, v, r) = \frac{r \Gamma(A) \Gamma(B)}{2\pi (uv)^{3/2}} \lim_{\delta \downarrow 0} {}_2F_1\left(A, B; 2; 1 - \frac{r^2 - (v - u - i\delta)^2}{4uv}\right). \quad (39)$$

Equations (32) and (39) equate limiting distributions. The two finite regulators need not represent the same function. An exact analytic displacement of the mode times would also change u, v and their products, rather than shifting only $v - u$.

This establishes the scalar two-point function using an existing transform evaluation. It does not constitute a direct integration of Eq. (32). Such a proof would be a useful mathematical supplement to the project; the Legendre conversion above only makes the application of the known result explicit.

A normalization check is available without special-function continuation. For $\nu = 1/2$,

$$H_{1/2}^{(1)}(z) = -i\sqrt{\frac{2}{\pi z}} e^{iz}, \quad H_{1/2}^{(2)}(z) = i\sqrt{\frac{2}{\pi z}} e^{-iz}. \quad (40)$$

At $u = v$ and $r > 0$, Eq. (32) reduces to

$$\frac{2}{\pi u} \lim_{\epsilon \downarrow 0} \int_0^\infty dk \sin(kr) e^{-\epsilon k} = \frac{2}{\pi ur}. \quad (41)$$

Equation (39) gives the same answer since $\Gamma(2)\Gamma(1) = 1$ and ${}_2F_1(2, 1; 2; s) = (1 - s)^{-1}$.

5 Checks and domain of the state

5.1 Invariant Klein–Gordon equation

The coordinate invariant satisfies

$$\nabla_\mu Z \nabla^\mu Z = H^2(1 - Z^2), \quad \square Z = -4H^2 Z. \quad (42)$$

For example, keeping x' fixed and writing $\eta = \eta_1$, $\eta' = \eta_2$,

$$\partial_\eta Z = \frac{\eta^2 - \eta'^2 + r^2}{2\eta^2\eta'}, \quad \partial_{\eta'}^2 Z = \frac{\eta'^2 - r^2}{\eta^3\eta'}, \quad \nabla^2 Z = -\frac{3}{\eta\eta'}. \quad (43)$$

Substitution into Eq. (7) gives

$$\square Z = H^2 \left(-\frac{2\eta'}{\eta} - \frac{2\eta}{\eta'} + \frac{2r^2}{\eta\eta'} \right) = -4H^2 Z. \quad (44)$$

The first identity follows by contracting the first derivatives with $g^{\mu\nu}$.

For a biscalar $G(Z)$, the homogeneous Klein–Gordon equation becomes

$$(1 - Z^2)G_{ZZ} - 4ZG_Z - \frac{M_\phi^2}{H^2}G = 0. \quad (45)$$

With $s = (1 + Z)/2$, it reads

$$s(1 - s)G_{ss} + (2 - 4s)G_s - \frac{M_\phi^2}{H^2}G = 0. \quad (46)$$

This is Gauss' equation with $c = 2$, $A + B = 3$ and

$$AB = \frac{9}{4} - \nu^2 = \frac{M_\phi^2}{H^2}, \quad (47)$$

as required by Eq. (36). The Wightman boundary value is a homogeneous bisolution, including in the distributional sense. Delta-function sources arise only after time ordering.

5.2 Short distance and the conformal limit

At coincidence $s \rightarrow 1$, and since $A + B = 3$,

$${}_2F_1(A, B; 2; s) \sim \frac{1}{\Gamma(A)\Gamma(B)(1 - s)}. \quad (48)$$

The gamma factors cancel, giving

$$G^+(x, x') \sim \frac{H^2}{4\pi^2 y_+} = \frac{1}{4\pi^2 a_1 a_2} \frac{1}{r^2 - (\Delta\eta - i0)^2}, \quad (49)$$

where $y_+ = \lim_{\delta \downarrow 0} 2(1 - Z_\delta)$. This agrees with the ultraviolet mode calculation. It is the leading Hadamard singularity; the mass and curvature also enter the subleading terms.

The first logarithm can be fixed directly from Eq. (45). In the y variable the equation is

$$(4y - y^2)G_{yy} + (8 - 4y)G_y - \frac{M_\phi^2}{H^2}G = 0. \quad (50)$$

Inserting $G = \alpha/y + \beta \ln(y/4) + \gamma + \dots$ gives $4\beta = (M_\phi^2/H^2 - 2)\alpha$. Thus, near coincidence,

$$G^+ = \frac{H^2}{4\pi^2 y_+} + \frac{M_\phi^2 - 2H^2}{16\pi^2} \ln\left(\frac{y_+}{4}\right) + C_{\text{BD}} + O(y_+ \ln y_+). \quad (51)$$

The finite constant is fixed by the state and parameters; changing the constant inside the logarithm changes its definition. The logarithmic coefficient is proportional to $m_\phi^2 + (\xi - 1/6)R$, as expected for a scalar Hadamard expansion.

For $m_\phi = 0$ and $\xi = 1/6$, one has $M_\phi^2 = 2H^2$, $\nu = 1/2$, $A = 2$, $B = 1$. Equation (36) then reduces exactly to

$$G_{\text{conf}}^+(x, x') = \frac{1}{4\pi^2 a_1 a_2} \frac{1}{r^2 - (\Delta\eta - i0)^2}. \quad (52)$$

Only in this case is the conformal-Minkowski expression the entire correlator, rather than its leading short-distance term. Dimensional analysis supplies a further check: ϕ has mass dimension one, and Eq. (36) has mass dimension two through its prefactor H^2 .

5.3 Infrared convergence and exceptional masses

For real $\nu > 0$, the small- k behavior is $H_\nu^{(1)}(-k\eta) \sim k^{-\nu}$. The equal-time radial measure therefore scales as

$$k^2 |\phi_k(\eta)|^2 dk \sim k^{2-2\nu} dk, \quad \int_0 dk k^{2-2\nu} < \infty \iff \nu < \frac{3}{2}. \quad (53)$$

At $\nu = 0$, logarithms replace the power and remain integrable. For imaginary order $\nu = i\rho$, the small- k factors oscillate in $\ln k$ with bounded magnitude at fixed ρ , and the $k^2 dk$ measure is also integrable. The condition for this principal-branch family is therefore $\text{Re } \nu < 3/2$.

At $M_\phi^2 = 0$, $\nu = 3/2$ and the integral develops a logarithmic infrared divergence, matching the pole $\Gamma(3/2 - \nu) = \Gamma(0)$ in the invariant formula. The exceptional locus is

$$m_\phi^2 + 12\xi H^2 = 0; \quad (54)$$

it is not restricted to the minimally coupled massless point unless $m_\phi^2 \geq 0$ and $\xi \geq 0$ are separately imposed. For $M_\phi^2 < 0$ the small-momentum divergence is stronger. Analytically continuing the hypergeometric expression to these masses does not produce an infrared-finite mode sum or establish a positive Bunch–Davies state. A separate infrared prescription, generally with additional state information, would be required before using such a case in the loop.

6 Boundary values and derivatives for the polarization

6.1 Closed-time-path components and contact equations

At $A_\mu = 0$, the charge-symmetric Bunch–Davies state gives

$$G^-(x, x') \equiv \langle \hat{\phi}^\dagger(x') \hat{\phi}(x) \rangle = G^+(x', x). \quad (55)$$

With all contour correlators defined without an additional factor of i ,

$$G^{-+}(x, x') = G^+(x, x'), \quad G^{+-}(x, x') = G^-(x, x'), \quad (56)$$

$$G^{++}(x, x') = \theta(\eta_1 - \eta_2) G^+(x, x') + \theta(\eta_2 - \eta_1) G^-(x, x'), \quad (57)$$

$$G^{--}(x, x') = \theta(\eta_2 - \eta_1) G^+(x, x') + \theta(\eta_1 - \eta_2) G^-(x, x'). \quad (58)$$

Here G^{++} is the time-ordered expectation value in the specified Bunch–Davies state. This definition alone does not identify it with a distinct in–out matrix element $\langle \text{out} | T \phi \phi^\dagger | \text{in} \rangle / \langle \text{out} | \text{in} \rangle$: that object requires an out-state prescription. For a causal expectation-value equation, the relevant response is constructed from the closed-time-path polarization and its retarded combination.

The delta-function equations fix a useful sign convention for the next calculation. Define $\delta_g(x, x')$ by

$$\int d^4 x' \sqrt{-g(x')} \delta_g(x, x') f(x') = f(x). \quad (59)$$

Differentiating the step functions in Eq. (57) and using Eq. (16) yields

$$(\square_x - M_\phi^2) G^{++}(x, x') = i \delta_g(x, x'), \quad (60)$$

$$(\square_x - M_\phi^2) G^{--}(x, x') = -i \delta_g(x, x'), \quad (61)$$

$$(\square_x - M_\phi^2) G^{-+}(x, x') = (\square_x - M_\phi^2) G^{+-}(x, x') = 0. \quad (62)$$

For example, the first-time-derivative jump in G^{++} is $-i a^{-2} \delta^{(3)}(\mathbf{x} - \mathbf{x}')$; the $-a^{-2} \partial_\eta^2$ term in $\square$ then gives $+i \delta_g$. Differentiation of a time-ordered correlator cannot therefore be replaced everywhere by ordinary differentiation of a smooth function of Z .

6.2 Invariant derivatives away from contact terms

For any smooth branch $G(Z)$,

$$\nabla_\mu G = G_Z \partial_\mu Z, \quad (63)$$

$$\nabla_\mu \nabla_{\nu'} G = G_{ZZ} (\partial_\mu Z) (\partial_{\nu'} Z) + G_Z \partial_\mu \partial_{\nu'} Z. \quad (64)$$

The primed index belongs to the tangent space at x' . For this mixed derivative of a biscalar there is no connection between the two points, so the partial derivatives in Eq. (64) are sufficient.

With $\Delta x_i = x_i - x'_i$, the coordinate ingredients are

$$\partial_{x_i} Z = -\frac{\Delta x_i}{\eta_1 \eta_2}, \quad \partial_{x'_j} Z = \frac{\Delta x_j}{\eta_1 \eta_2}, \quad \partial_{x_i} \partial_{x'_j} Z = \frac{\delta_{ij}}{\eta_1 \eta_2}, \quad (65)$$

$$\partial_{\eta_1} Z = \frac{\eta_1^2 - \eta_2^2 + r^2}{2\eta_1^2 \eta_2}, \quad \partial_{\eta_2} Z = \frac{\eta_2^2 - \eta_1^2 + r^2}{2\eta_1 \eta_2^2}. \quad (66)$$

The remaining mixed components are

$$\partial_{\eta_1} \partial_{\eta_2} Z = -\frac{\eta_1^2 + \eta_2^2 + r^2}{2\eta_1^2 \eta_2^2}, \quad (67)$$

$$\partial_{\eta_1} \partial_{x'_j} Z = -\frac{\Delta x_j}{\eta_1^2 \eta_2}, \quad \partial_{x_i} \partial_{\eta_2} Z = \frac{\Delta x_i}{\eta_1 \eta_2^2}. \quad (68)$$

For $s = (1 + Z)/2$ and the constants in Eq. (35),

$$G_Z = C_\nu \frac{AB}{4} {}_2F_1(A+1, B+1; 3; s), \quad (69)$$

$$G_{ZZ} = C_\nu \frac{AB(A+1)(B+1)}{24} {}_2F_1(A+2, B+2; 4; s). \quad (70)$$

These follow from $d {}_2F_1(a, b; c; s)/ds = (ab/c) {}_2F_1(a+1, b+1; c+1; s)$ and $ds/dZ = 1/2$. In particular,

$$G_Z = \frac{M_\phi^2}{64\pi^2} \Gamma(A) \Gamma(B) {}_2F_1(A+1, B+1; 3; s). \quad (71)$$

The contour prescription must be retained when taking boundary values of these derivatives.

7 What this supplies to the next calculation

The free scalar problem supplies the normalized mode functions, the invariant two-point distribution in its convergent mass domain, and the contour and derivative identities above. The normalization, Klein–Gordon equation, local singularity and conformal limit agree. The only step imported as a substantive special-function result is the evaluation of the full mode sum.

For the polarization, these formulas must be combined with the two derivative vertices, the seagull vertex and the contact terms from time ordering. Their sum, after a consistent extension at coincidence, is the object to test against the Ward identity. The local logarithmic divergence can then be compared with the heat-kernel calculation in Step 2. This scalar input does not fix the finite part of δ_3 : that requires a renormalization prescription for the chosen response, with its state and boundary conditions specified. The free vector analysis in Step 3 concerns that response’s external modes; no vector propagator is an internal line of the scalar one-loop 1PI diagram.

References

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